Group I, Row V, part 2b (UV-finite part): plus prescription

1 Result

- **Exact.** The plus prescription is done on $\Lambda < p^+ < k^+ - \Lambda$, the range of your expression. No limit $\Lambda \to 0$ is taken. The only identity used is

$$\int_\Lambda^{k^+-\Lambda} \frac{dp^+}{p^+} = \int_\Lambda^{k^+-\Lambda} \frac{dp^+}{k^+ - p^+} = \ln \frac{k^+ - \Lambda}{\Lambda} = \log \frac{\vee}{\Lambda} - \log \frac{\vee}{k^+} + \log\Big(1 - \frac{\Lambda}{k^+}\Big). \tag{1}$$

- **Two singular points.** There are only two:
  - $p^+ \to 0$: the $\frac{k^+-p^+}{p^+}$ term of the bracket, and the $\frac{k^+-p^+}{p^+}$ of the counterterm;
  - $p^+ \to k^+$: the $\frac{p^+}{k^+-p^+}$ term of the bracket, and the $\frac{p^+}{k^+-p^+}$ of the counterterm.

  They are subtracted at $p^+ = 0$ and $p^+ = k^+$. The subtractions are put back as endpoint terms multiplying $\ln \frac{k^+-\Lambda}{\Lambda}$, Eq. (11).

- **Counterterm.** It depends on p^+ only through $\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}$. Its $\frac{1}{p^+}$ and $\frac{1}{k^+-p^+}$ parts therefore go entirely into the endpoint terms, and only $\frac{p^+(k^+-p^+)}{k^{+2}} - 2$ stays under the p^+ integral.

- **UV finite piece by piece.** The plus-prescribed integral and both endpoint terms are UV finite ($x \to z$) separately (Sec. 5).

- **Large distances.** The original converges at large $|x - z|$ because of the phase $e^{-ik\cdot(z-x)p^+/k^+}$. The endpoint terms sit at $p^+ = 0$ and $p^+ = k^+$, where this phase no longer cuts off large distances. Separately, the endpoint terms and the plus-prescribed integral therefore fall only like $1/(x-z)^2$, and their transverse integrals are logarithmically divergent at large distance. The two divergences cancel in the sum. The split is exact at fixed transverse points; transverse integrals must be done for the sum, or with the same large-distance cutoff in both parts (Sec. 5).

- **Checked numerically** at random transverse points (Sec. 6):
  - plus-prescribed integral + endpoint terms = your p^+ integral, to 10^{-15};
  - the endpoint coefficients agree with the limits $p^+ \to 0$, $p^+ \to k^+$ of your integrand.

2 The expression

As sent; $\int_{x,y,z,x',w'}$ is over all transverse points.

$$\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{ \\
e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}} - \frac{p^+}{k^+-p^+}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right] \\
+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right] \\
+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]
\end{aligned}$$

$$
- \left[\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]
$$

$$
\times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)
$$

$$
- \frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}
$$

$$
\times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\}. \tag{2}
$$

Below, $D \equiv p^+(y-x)^2 + (k^+-p^+)(y-z)^2$, and the two colour operators are

$$
\mathcal{O}_x \equiv \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y),
$$

$$
\mathcal{O}_z \equiv \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y). \tag{3}
$$

3 Plus prescription on $\Lambda < p^+ < k^+ - \Lambda$

For any function $f(p^+)$ that is smooth on $0 \le p^+ \le k^+$ define

$$
\int_\Lambda^{k^+-\Lambda} dp^+ \Big[\frac{k^+}{p^+}\Big]_+ f(p^+) \equiv \int_\Lambda^{k^+-\Lambda} dp^+\, \frac{k^+}{p^+}\big[f(p^+) - f(0)\big],
$$

$$
\int_\Lambda^{k^+-\Lambda} dp^+ \Big[\frac{k^+}{k^+-p^+}\Big]_+ f(p^+) \equiv \int_\Lambda^{k^+-\Lambda} dp^+\, \frac{k^+}{k^+-p^+}\big[f(p^+) - f(k^+)\big]. \tag{4}
$$

By Eq. (1), exactly,

$$
\int_\Lambda^{k^+-\Lambda} dp^+\, \frac{k^+}{p^+} f(p^+) = \int_\Lambda^{k^+-\Lambda} dp^+ \Big[\frac{k^+}{p^+}\Big]_+ f(p^+) + k^+ f(0)\ln\frac{k^+-\Lambda}{\Lambda},
$$

$$
\int_\Lambda^{k^+-\Lambda} dp^+\, \frac{k^+}{k^+-p^+} f(p^+) = \int_\Lambda^{k^+-\Lambda} dp^+ \Big[\frac{k^+}{k^+-p^+}\Big]_+ f(p^+) + k^+ f(k^+)\ln\frac{k^+-\Lambda}{\Lambda}. \tag{5}
$$

In what follows the bracket $[\,\cdot\,]_+$ acts on the whole p^+ dependence of the term it stands in: the phase $e^{-ik\cdot(z-x)p^+/k^+}$, $1/D$ and the brackets. Acting on a function that does not depend on p^+, it gives zero. We write the singular coefficients as

$$
-\frac{k^+-p^+}{p^+} = 1 - \frac{k^+}{p^+}, \qquad -\frac{p^+}{k^+-p^+} = 1 - \frac{k^+}{k^+-p^+},
$$

$$
\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+} = \frac{p^+(k^+-p^+)}{k^{+2}} - 2 + \frac{k^+}{p^+} + \frac{k^+}{k^+-p^+}, \tag{6}
$$

and apply Eq. (5) to every $\frac{k^+}{p^+}$ and $\frac{k^+}{k^+-p^+}$.

4 The result

$$
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \frac{d^3N_{2b}}{d^3k}\bigg|_{\text{plus}} + \frac{d^3N_{2b}}{d^3k}\bigg|_{\text{endpoint}}. \tag{7}
$$

4.1 The plus-prescribed integral

Changes with respect to your expression are in blue.

$$
\frac{d^3N_{2b}}{d^3k}\bigg|_{\text{plus}} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{
$$

$$
e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2 + (k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}
$$

$$
\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\big[(y-x)^2 - (y-z)^2\big]\,\frac{p^+(k^+-p^+)}{k^{+2}}
$$

$$
\begin{aligned}
&+\left(1-\Big[\frac{k^+}{k^+-p^+}\Big]_+\right)\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&+\left(1-\Big[\frac{k^+}{p^+}\Big]_+\right)\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)\\
&-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}-2\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\}. \qquad (8)
\end{aligned}
$$

Written out, the two plus-prescribed terms are

$$
\begin{aligned}
-\Big[\frac{k^+}{k^+-p^+}\Big]_+F_2 &\equiv -\frac{k^+}{k^+-p^+}\big[F_2(p^+)-F_2(k^+)\big],\\
-\Big[\frac{k^+}{p^+}\Big]_+F_4 &\equiv -\frac{k^+}{p^+}\big[F_4(p^+)-F_4(0)\big], \qquad (9)
\end{aligned}
$$

with

$$
\begin{aligned}
F_2(p^+) &= e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{D}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\mathcal{O}_x,\\
F_4(p^+) &= e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{D}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\mathcal{O}_x, \qquad (10)
\end{aligned}
$$

and $F_2(k^+)$, $F_4(0)$ their values at $p^+=k^+$ ($D=k^+(y-x)^2$, phase $e^{-ik\cdot(z-x)}$) and $p^+=0$ ($D=k^+(y-z)^2$, phase 1).

4.2 The endpoint terms

With $\frac{1}{k^+}\int\frac{dp^+}{2\pi}k^+=\frac{1}{2\pi}$, $\frac{g^4}{8\pi^4}\frac{1}{2\pi}=\frac{g^4}{16\pi^5}$, and the contractions done:

$$
\begin{aligned}
\left.\frac{d^3N_{2b}}{d^3k}\right|_{\text{endpoint}} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\ln\Big(\frac{k^+-\Lambda}{\Lambda}\Big)\int_{x,y,z,x',w'}\Bigg\{\\
&(p^+\to 0)\quad e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\Bigg[\Big(\frac{1}{(x-z)^2}-\frac{(y-x)\cdot(z-x)}{2(y-x)^2(z-x)^2}\Big)\mathcal{O}_x\\
&\qquad\qquad -\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\,\mathcal{O}_z\Bigg]\\
&+(p^+\to k^+)\quad e^{-ik\cdot(w'-x)}\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\Big(\frac{1}{(x-z)^2}-\frac{(y-z)\cdot(x-z)}{2(y-z)^2(x-z)^2}\Big)\mathcal{O}_x\\
&\qquad -e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\,\mathcal{O}_z\Bigg\}. \qquad (11)
\end{aligned}
$$

In each line the $\mathcal{O}_x$ term comes from the bracket of your expression, and the $\mathcal{O}_z$ term from the counterterm (half of its $2\ln\frac{k^+-\Lambda}{\Lambda}$ in each line). The labels mark the endpoint each line comes from. In the notation $K(a,b,c)=\frac{(a-c)\cdot(b-c)}{(a-c)^2(b-c)^2}$ of your notes, the kernels are $K(z,z,x)-\frac{1}{2}K(y,z,x)$, $K(x,x,z)-\frac{1}{2}K(y,x,z)$ and $K(z,z,x)-K(y,z,x)$.

Combined form. Rename $x\leftrightarrow z$ in the $\mathcal{O}_x$ term of the second line, which is allowed under $\int_{x,z}$. Then all terms have the phase $e^{-ik\cdot(w'-z)}$, i.e. the measured gluon at z:

$$
\left.\frac{d^3N_{2b}}{d^3k}\right|_{\text{endpoint}} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\ln\Big(\frac{k^+-\Lambda}{\Lambda}\Big)\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}
$$

$$\times N_c\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Bigg\{\frac{1}{(x-z)^2}\Big[U^{af}(z) - U^{af}(x)\Big] + \frac{(y-x)\cdot(z-x)}{(y-x)^2(z-x)^2}\Big[U^{af}(y) - \tfrac{3}{2}\,U^{af}(z) + \tfrac{1}{2}\,U^{af}(x)\Big]\Bigg\}\rho^{b'}(x')\,\rho^{f}(y). \tag{12}$$

With Eq. (1), $\ln\frac{k^+-\Lambda}{\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log(1-\frac{\Lambda}{k^+})$, so Eq. (11) gives a $\log\frac{\vee}{\Lambda}$ term, a $\log\frac{\vee}{k^+}$ term and a finite $\log(1-\frac{\Lambda}{k^+})$ term, all with the same transverse integral.

5 Properties

- **UV ($x \to z$).** Your expression is UV finite at every p^+: the $\frac{1}{(x-z)^2}$ terms of the bracket and of the counterterm cancel. After the split, each piece is UV finite by itself.
  - In $F(p^+) - F(0)$ and $F(p^+) - F(k^+)$ the leading $\frac{1}{(x-z)^2}$ term does not depend on p^+ (at $x = z$ the phase is 1 and $D = k^+(y-z)^2$), so it drops out.
  - The remaining p^+ factors of the bracket are $\frac{p^+(k^+-p^+)}{k^{+2}}$, 1, $\frac{k^+-p^+}{k^+}$, 1, $\frac{p^+}{k^+}$, -1. With the angular averages of Part2b_MSbar.pdf, Sec. 3, the coefficient of $\frac{1}{(x-z)^2}$ is $\frac{p^+(k^+-p^+)}{k^{+2}} - 2$, independent of $d_\perp$. It cancels against the counterterm, which keeps exactly $\frac{p^+(k^+-p^+)}{k^{+2}} - 2$.
  - In the endpoint terms the $\mathcal{O}_x$ and $\mathcal{O}_z$ terms cancel at $x \to z$, line by line. In the combined form (12), $\frac{1}{(x-z)^2}$ multiplies $U(z) - U(x) = O(|x-z|)$.

  Numerically, $r^2\langle\cdot\rangle_{\text{angle}} \propto r^2$ for every piece, while it tends to a constant for 2b before the counterterm (Table 1).

- **Large distances.** At fixed p^+ your integrand falls off at large $|x-z|$ through the phase $e^{-ik\cdot(z-x)p^+/k^+}$. The endpoint terms do not have this cutoff:
  - at $p^+ = 0$ the phase is 1, and the $\mathcal{O}_x$ kernel falls like $\frac{1}{2(x-z)^2}$;
  - at $p^+ = k^+$ the $\mathcal{O}_x$ term does not depend on the phase of z, and falls like $\frac{1}{2(x-z)^2}$ in z.

  $\mathcal{O}_x$ does not vanish at large distance ($U \to 1$ outside the target). So the x integral of the $p^+ \to 0$ term and the z integral of the $p^+ \to k^+$ term are logarithmically divergent at large distance. The plus-prescribed integrand has the same divergence with the opposite sign, from the subtracted $F_4(0)$ and $F_2(k^+)$, and the sum converges. Numerically, $r^2\langle\cdot\rangle$ at large r tends to a constant for the endpoint terms and for the plus-subtracted term, and decreases for your integrand (Table 2).

  So the split (8)+(11) is exact pointwise in the transverse coordinates. The transverse integrals must be taken of the sum, or with the same large-distance cutoff in both parts, before they are separated. The $\log\frac{\vee}{\Lambda}$ part of Eq. (11) is part of the evolution terms; there such large-distance pieces combine with the other rows.

6 Checks

Script `plus2b_checks.py` (and `plus2b_ir.py` for the large-distance behaviour). Random transverse points; $k^+ = 1.7$, $\Lambda = 0.11$, $d_\perp = 2$. The colour operator is modelled by a smooth complex function $\mathcal{O}(p)$ of the point p where the Wilson line of the charge at y sits: $\mathcal{O}_x = \mathcal{O}(x)$, $\mathcal{O}_z = \mathcal{O}(z)$.

Table 1: UV and exactness.

check	result
config 1: endpoint coefficients vs. limits of the integrand at $p^+ \to 0$, k^+	rel. 7×10^{-7}, 2×10^{-8} (limit taken at 10^{-8})
config 1: plus-prescribed integrand of Eq. (8) vs. its definition	rel. 6×10^{-16}
config 1: your p^+ integral vs. plus + endpoint	$-0.9827150403 + 0.1218313774\,i$ vs. $-0.9827150403 + 0.1218313774\,i$ (rel. 4×10^{-16})
config 2: endpoint coefficients vs. limits of the integrand at $p^+ \to 0$, k^+	rel. 3×10^{-8}, 6×10^{-8} (limit taken at 10^{-8})
config 2: plus-prescribed integrand of Eq. (8) vs. its definition	rel. 4×10^{-16}
config 2: your p^+ integral vs. plus + endpoint	$0.0668557877 - 0.0373400904\,i$ vs. $0.0668557877 - 0.0373400904\,i$ (rel. 2×10^{-15})

config 3: endpoint coefficients vs. limits of the integrand at $p^+ \to 0$, k^+	rel. 2×10^{-7}, 2×10^{-8} (limit taken at 10^{-8})
config 3: plus-prescribed integrand of Eq. (8) vs. its definition	rel. 7×10^{-18}
config 3: your p^+ integral vs. plus + endpoint	$0.0023663954 - 0.0028823188\,i$ vs. $0.0023663954 - 0.0028823188\,i$ (rel. 6×10^{-16})
UV: $r^2\langle\cdot\rangle$, 2b as written (fixed ξ), $r = 10^{-2}, 10^{-3}, 10^{-4}$	6.34×10^{-3}, 6.35×10^{-3}, 6.35×10^{-3}
UV: $r^2\langle\cdot\rangle$, rest (fixed ξ), $r = 10^{-2}, 10^{-3}, 10^{-4}$	1.64×10^{-5}, 1.64×10^{-7}, 1.64×10^{-9}
UV: $r^2\langle\cdot\rangle$, plus-prescribed integrand (fixed ξ), $r = 10^{-2}, 10^{-3}, 10^{-4}$	7.43×10^{-6}, 7.43×10^{-8}, 7.43×10^{-10}
UV: $r^2\langle\cdot\rangle$, endpoint $p^+ \to 0$ (I_0), $r = 10^{-2}, 10^{-3}, 10^{-4}$	6.73×10^{-8}, 6.73×10^{-10}, 6.73×10^{-12}
UV: $r^2\langle\cdot\rangle$, endpoint $p^+ \to k^+$ (I_1), $r = 10^{-2}, 10^{-3}, 10^{-4}$	1.50×10^{-5}, 1.50×10^{-7}, 1.50×10^{-9}
combined form (12) vs. Eq. (11), x, z summed on a grid with a symmetric weight	rel. 1×10^{-13}

Table 2: Large distances: $r^2\langle\cdot\rangle_{\text{angle}}$ at $r = 10, 100, 1000$, with $\mathcal{O}(p) \to$ const at large $|p|$.

piece	$r^2\langle\cdot\rangle$
original integrand, $\xi = 0.37$ (x far)	2.375×10^{-2}, 4.755×10^{-3}, 3.822×10^{-4}
endpoint $p^+ \to 0$, I_0 (x far)	2.739×10^{-3}, 2.739×10^{-3}, 2.739×10^{-3}
plus part, -($F_4(\xi)$-$F_4(0)$)/ξ at $\xi = 0.37$ (x far)	7.439×10^{-3}, 7.403×10^{-3}, 7.403×10^{-3}
endpoint $p^+ \to k^+$, I_1 (z far)	6.657×10^{-2}, 1.008×10^{-1}, 1.019×10^{-1}